

FuzzOrigin: Detecting UXSS vulnerabilities in Browsers through Origin Fuzzing

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FuzzOrigin: Detecting UXSS vulnerabilities in Browsers through Origin Fuzzing

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Universal cross-site scripting (UXSS) is a browser vulnerability, making a vulnerable browser execute an attacker's script on any web pages loaded by the browser. UXSS is considered a far more severe vulnerability than well-studied cross-site scripting (XSS). This is because the impact of UXSS is not limited to a web application, but it impacts each and every web application as long as a victim user runs a vulnerable browser. We find that UXSS vulnerabilities are difficult to find, especially through fuzzing, for the following two reasons. First, it is challenging to detect UXSS because it is a semantic vulnerability. In order to detect UXSS, one needs to understand the complex interaction semantics between web pages. Second, it is difficult to generate HTML inputs that trigger UXSS since one needs to drive the browser to perform complex interactions and navigations.

This paper proposes FuzzOrigin, a browser fuzzer designed to detect UXSS vulnerabilities. FuzzOrigin addresses the above two challenges by (i) designing an origin sanitizer with a static origin tagging mechanism and (ii) prioritizing origin-update operations through generating chained-navigation operations handling dedicated events. We implemented FuzzOrigin, which works with most modern browsers, including Chrome, Firefox, Edge, and Safari. During the evaluation, FuzzOrigin discovered four previously unknown UXSS vulnerabilities, one in Chrome and three in Firefox, all of which have been confirmed by the vendors. FuzzOrigin is responsible

for finding one out of two UXSS vulnerabilities in Chrome reported in 2021 and all three in Firefox, highlighting its strong effectiveness in finding new UXSS vulnerabilities.

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